

AKHIL JAINI

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SUMMARY

Computational Scientist and PhD Researcher with 6+ years of experience designing scalable real-time data-processing pipelines, high-performance software, and predictive statistical modelling for large-scale datasets. Bringing expertise in machine learning, Python and SQL analytics, and scientific computing, with a proven track record of extracting actionable insights from complex observational datasets.

EDUCATION

PhD in Computational Astrophysics	2023-Ongoing
Swinburne University of Technology, Melbourne, Australia	
Master of Technology in Astronomical Instrumentation	2020-2022
Indian Institute of Astrophysics, Bangalore, India and University of Calcutta, Kolkata, India	
Bachelor of Engineering (Honours) in Electrical and Electronics	2014-2018
Birla Institute of Technology and Science – Pilani, Pilani, India	

TECHNICAL SKILLS

Programming Languages: Python, C, C++, CUDA C, SQL, Rust, HTML, JavaScript, TypeScript

Machine Learning: PyTorch, Scikit-Learn, TensorFlow, NumPy, SciPy, Pandas

Software Engineering: FastAPI, REST APIs, Docker, Git, GitHub Actions

Data & Infrastructure: PostgreSQL, SQLite, Linux (Bash/Shell), AWS, HPC (OzStar/Slurm)

RESEARCH IMPACT

First author of multiple peer-reviewed publications developing scalable computational methods and software adopted across international astronomy collaborations. **Full list of Publications:** ui.adsabs.harvard.edu/public-libraries/6AWkqcbzTQWau_usyl5V8Q

EXPERIENCE

PhD Researcher – Centre for Astrophysics and Supercomputing, Swinburne University, Melbourne, VIC.	June 2023-Present
- Designed and developed predictive modelling and spatial cross-matching algorithms across multi-terabyte datasets using Python and SQL, improving feature localisation precision by 4x from ~1 to <0.25 arcseconds and enabling reliable downstream analysis. - Built and deployed a real-time data processing pipeline for ingesting and processing streaming sensor data at >20 TB/s throughput, developing anomaly detection algorithms to identify millisecond-scale extragalactic events. - Characterised and modelled complex time-series and polarimetric data using custom Python, SQL and statistical methods. - Collaborated within multidisciplinary international teams to translate complex analytics into production-grade astronomy software.	
Research Software Engineer – Indian Institute of Astrophysics, Bangalore, India.	August 2021-April 2023
- Designed and validated data analysis workflows to identify and correct systematic errors in high-precision observational datasets, improving data quality and measurement reliability. - Developed Python and C++ control interfaces and data extraction pipelines to automate instrument calibration and control, high-precision spectrophotometric data acquisition, and end-user telemetry monitoring for a state-of-the-art space-based ultraviolet telescope.	
Computational Research Assistant – Raman Research Institute, Bangalore, India.	January 2018-March 2020
Designed and implemented a full-scale Python simulation framework to model multi-variable sensor networks for an orbiting radio interferometer, performing predictive risk analysis and scenario modelling under the rigorous operating conditions of outer space.	
Software Engineering Intern – National Centre for Radio Astrophysics, Pune, India.	May 2017-July 2017
Developed a scalable GPU-accelerated high-throughput signal processing pipeline using CUDA C for real-time astronomical event detection and validated performance using simulated streaming data.	

PROJECTS

- AstroConvert** (Python, Flet, AstroPy): Engineered a cross-platform application that intelligently detects, validates, and transforms heterogeneous scientific coordinate formats, reducing manual preprocessing through automated input recognition.
- AI-Powered Bushfire Early Warning System** (PyTorch, FastAPI, Docker): Built an end-to-end predictive risk classification system analysing Sentinel-1 radar imagery to detect active fire fronts under cloud and smoke, deployed via FastAPI with an interactive dashboard.
- Agentic Code Review & Document Analyst** (LangGraph, Ollama, SQLite): Developed a multi-agent AI system that autonomously analyses source code and technical documents, executes automated validation workflows, and generates structured review reports.